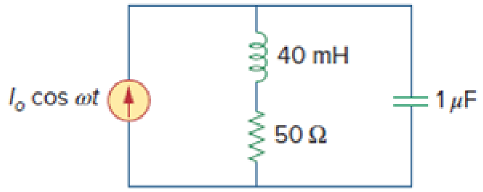
Chapter 14, Problem 39P

Problem

For the "tank" circuit in Fig. 14.79, find the resonant frequency.

Figure 14.79:



Step-by-step solution

Step 1 of 3

Refer to Figure 14.79 in the textbook.

The impedance of inductor is $j\omega L$.

The impedance of capacitor is $\frac{1}{j\omega C}$.

Redraw the circuit with their impedance values as shown in Figure 1.

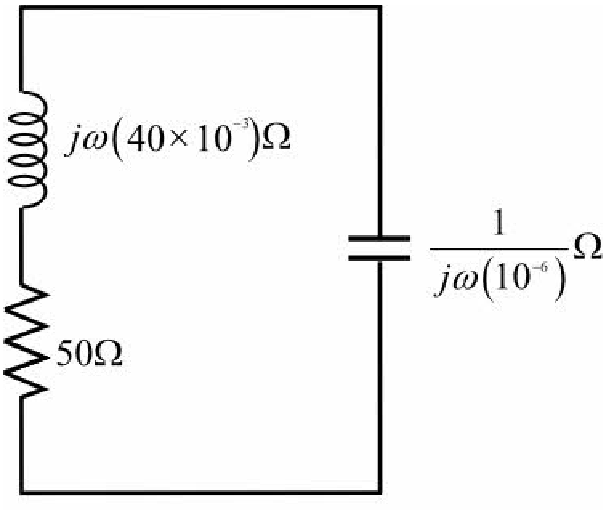


Figure 1

Step 2 of 3

Calculate the input admittance.

$$\begin{aligned}
\mathbf{Y} &= \frac{1}{50+j\omega(40\times10^{-3})}+\frac{1}{\frac{1}{j\omega(10^{-6})}} \\
&= \frac{1}{50+j\omega(40\times10^{-3})}+j\omega(10^{-6}) \\
&= \frac{50-j\omega(40\times10^{-3})}{(50)^2+(40\omega\times10^{-3})^2}+j\omega(10^{-6})
\end{aligned}$$

Simplify further.

$$\mathbf{Y} = \frac{50}{(50)^2 + \left(40\omega \times 10^{-3}\right)^2} - \frac{j\omega\left(40\times 10^{-3}\right)}{(50)^2 + \left(40\omega \times 10^{-3}\right)^2} + j\omega\left(10^{-6}\right)$$
$$= \frac{50}{(50)^2 + \left(40\omega \times 10^{-3}\right)^2} + j\left(10^{-6}\omega - \frac{\omega\left(40\times 10^{-3}\right)}{(50)^2 + \left(40\omega \times 10^{-3}\right)^2}\right)$$

At resonance imaginary part of **Y** is zero.

$$\text{Im}(\mathbf{Y}) = 0$$

$$10^{-6}\omega_0 - \frac{\omega_0\left(40\times 10^{-3}\right)}{(50)^2 + \left(40\omega_0^{\,2}\times 10^{-3}\right)^2} = 0$$

$$10^{-3} - \frac{(40)}{(50)^2 + \left(40\omega_0^{\,2}\times 10^{-3}\right)^2} = 0$$

$$\left((50)^2 + \left(40\omega_0^{\,2}\times 10^{-3}\right)^2\right)10^{-3} - 40 = 0$$

Simplify further.

$$2.5 + 16\times 10^{-7}\omega_0^{\,2} = 40$$
$$16\times 10^{-7}\omega_0^{\,2} = 37.5$$
$$\omega_0 = 4.841\frac{\text{krad}}{\text{s}}$$

Therefore, the resonant frequency is $\boxed{4.841\frac{\text{krad}}{\text{s}}}$.